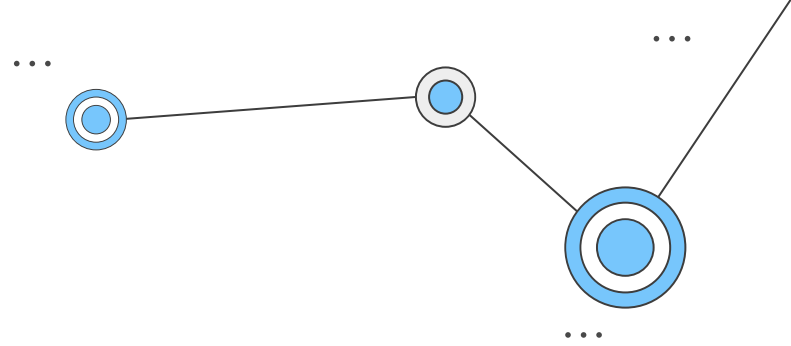
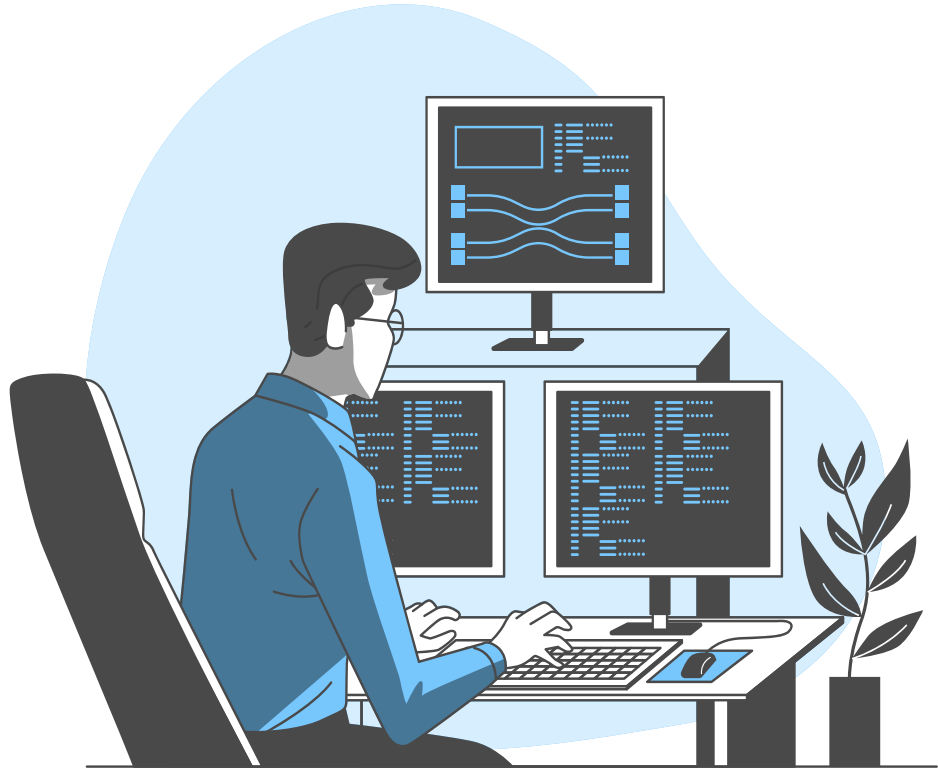
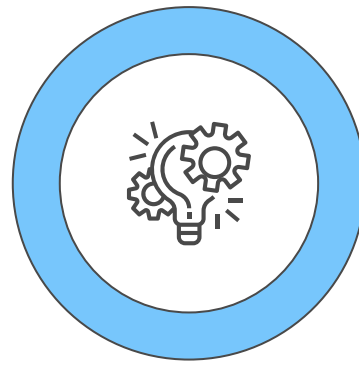




Cloud Computing



Topics

1st What is Cloud Computing

2nd Types of Cloud Computing

3rd Advantages and Disadvantages

4th Cloud vs local servers

6th Major Cloud Services Providers

5th Cloud Security

6th Major Cloud Services Providers

7th IoT in Cloud Computing and everyday use

8th How companies use and benefit from Cloud Services

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01

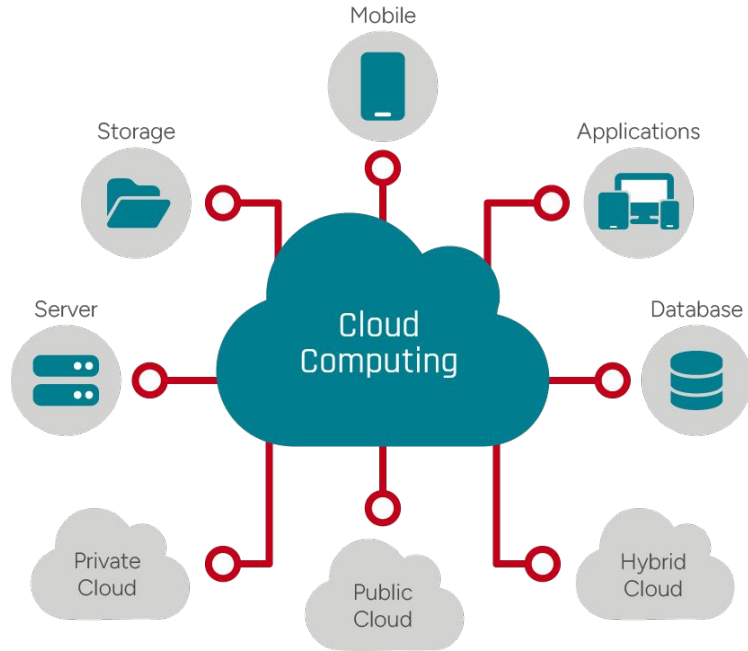
What is Cloud Computing?



What is Cloud Computing?

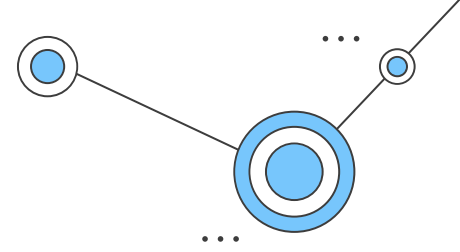
The on-demand availability of computing resources (such as storage and infrastructure), as services over the internet, instead of using your own local computer or physical servers.

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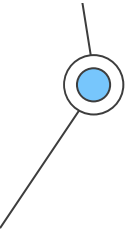
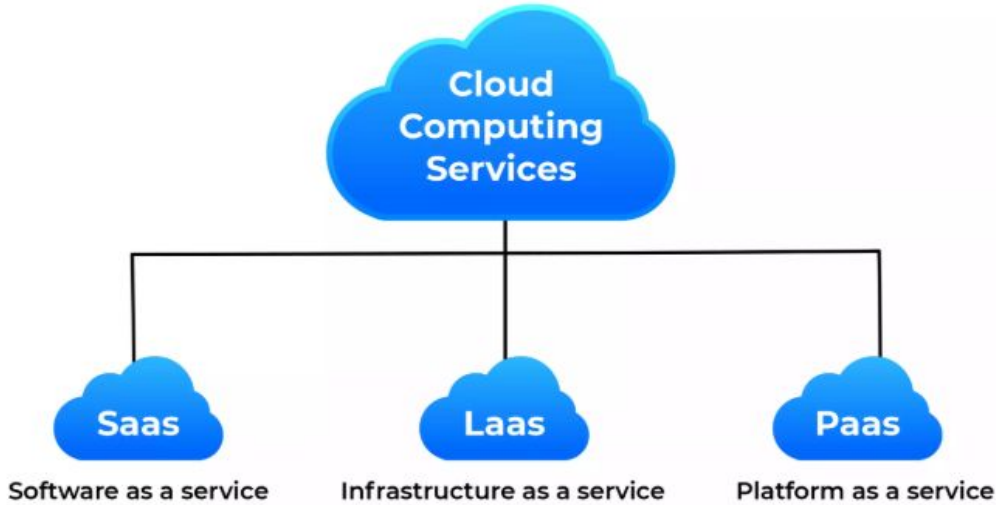
Examples:

- Using Google Drive to store files online;
- Saving photos to iCloud;
- Running websites without owning physical servers.

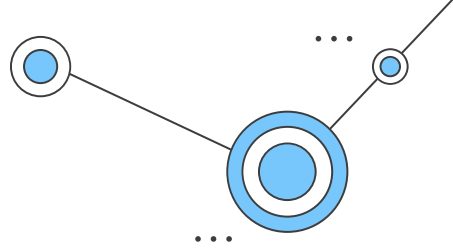


02

Types of Cloud Computing



Main types of cloud services



Infrastructure as a Service (IaaS)

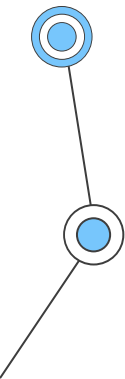
You rent virtual machines, storage, and networks

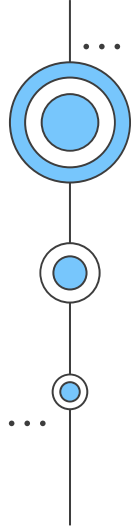
Software as a Service (SaaS)

You use software through a browser or app

Platform as a Service (PaaS)

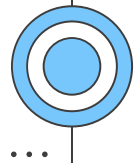
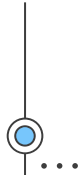
You deploy apps without managing hardware

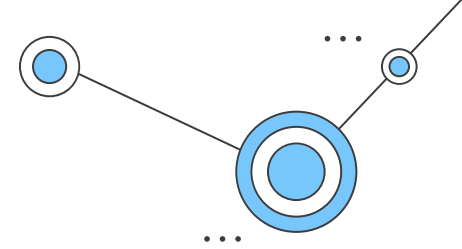




Types of Cloud Environments

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Public Cloud

Infrastructure is owned by a cloud provider and shared among many customers.

- lower cost

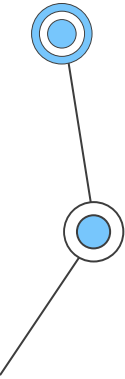
Private Cloud

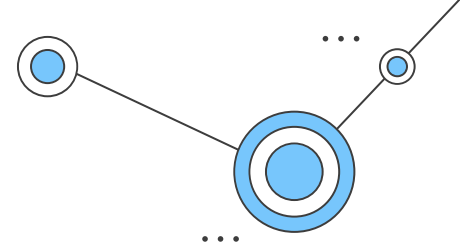
Infrastructure is used only by one organization.

- more control and security

Hybrid Cloud

A combination of the public and private clouds.





03

Advantages and Disadvantages of Cloud Computing

Key Benefits of Cloud Computing

- Cost Efficiency
- Scalability and Flexibility
- Accessibility and Remote collaboration
- Security and compliance
- Disaster Recovery
- Business Continuity

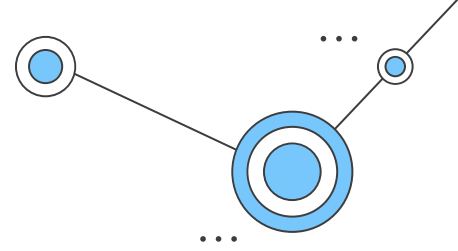


Cloud services advantages are significant, it can be a way for companies to **reduce their carbon footprint by as much as 30%**

Disadvantages of Cloud Computing

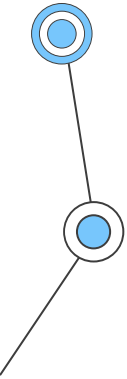
- ❖ Ongoing operational costs
- ❖ Security risks and Data Privacy concerns
- ❖ Reliability issues (performance issues, dependent on network connection)
- ❖ Limited control over infrastructure





04

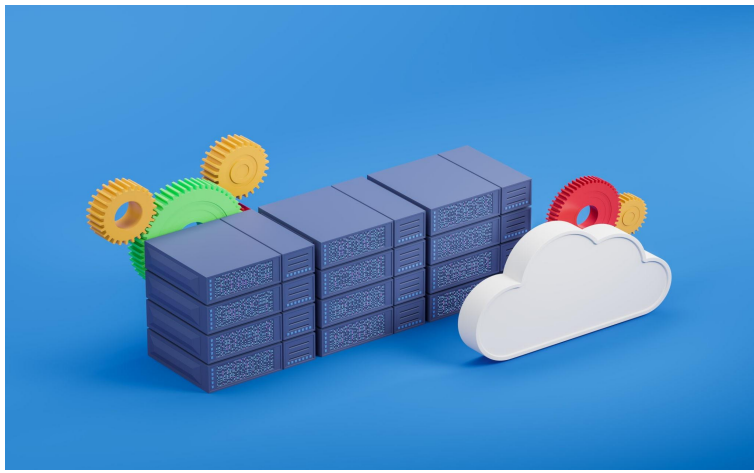
Cloud vs Local servers



Local (or on-premise) Servers

- ★ Enhanced Control over infrastructure and company data.
- ★ Performance not all-relying on Internet connection.
- ★ Require a significant upfront investment. Lower ongoing subscription costs.
- ★ Personalized Security measures, maintenance is of the company's responsibility.





Cloud Servers, and a Hybrid approach

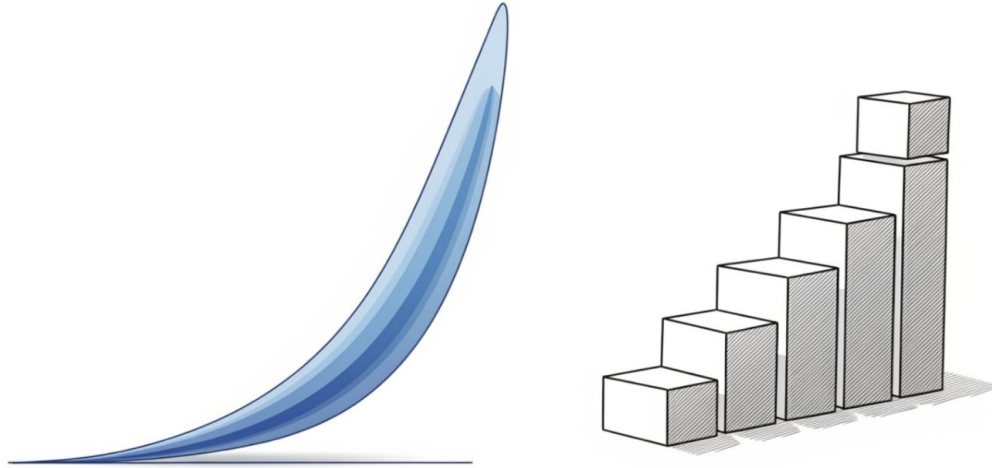
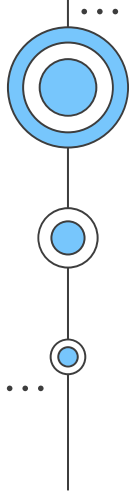
Remote access to the software from anywhere (with Internet connection)

Can easily scale up, more adaptable

Cloud provider is responsible for server maintenance

Subscription based model

**Critical workloads can stay local, but still needs scalability – better to go for a
Hybrid Approach**

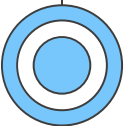


How to choose?

There is no right answer, and the answer can change over time.

It depends on funds available and the compromises each company is willing to make.

Most importantly: **make and informed decision**, weigh the pros and cons, and plan for future changes and the business evolution.





05 Cloud Security

What is it?

Is it important?

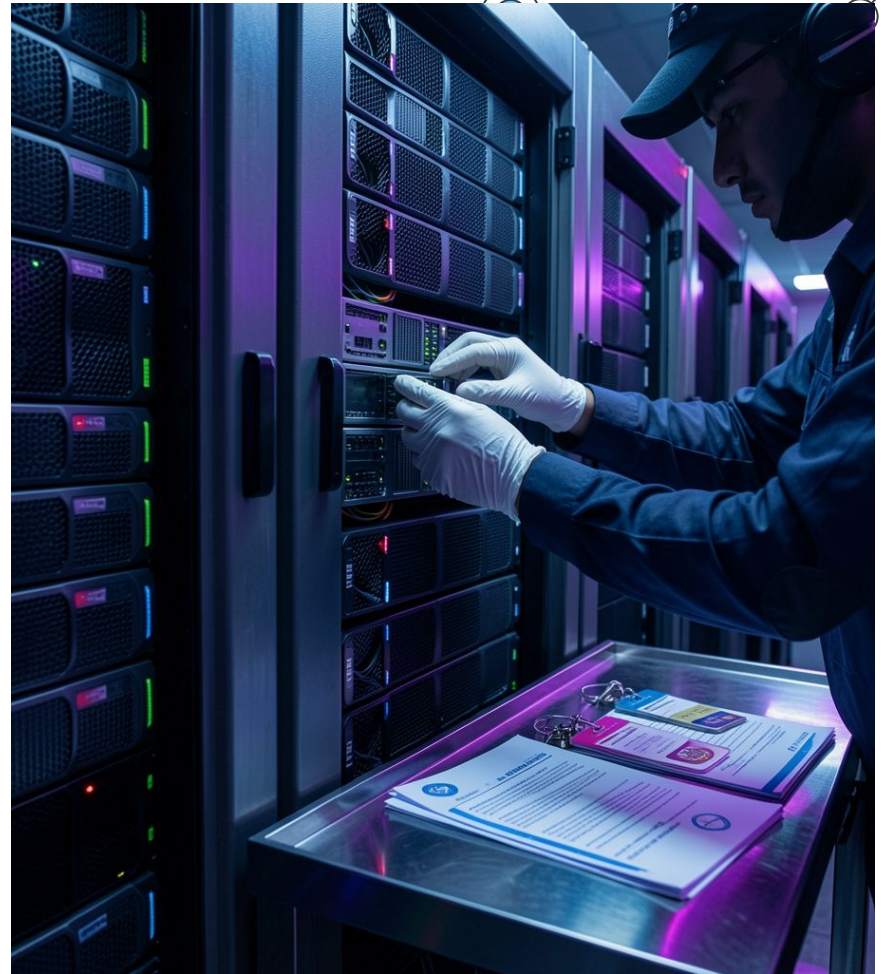
Main Security Threats

Data Breaches

Malware Attacks

Insider Threats

DDos Attacks



Security Solutions



Encryption

Multi-Factor Authentication

Firewalls

Identity and Access Management

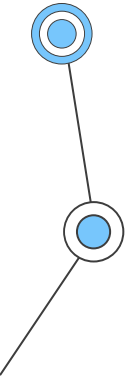
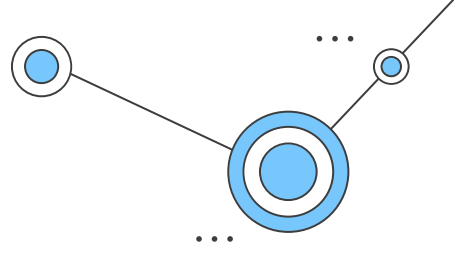
Regular Backups

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06

Major Cloud Services Providers

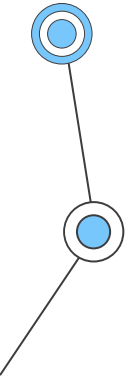
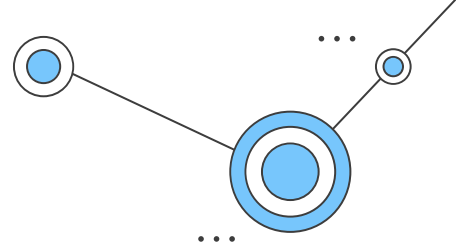


Cloud Services

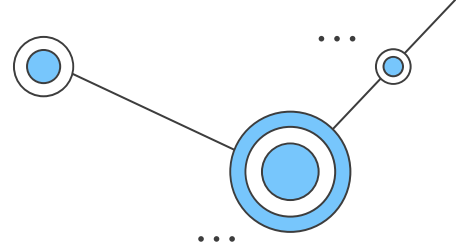
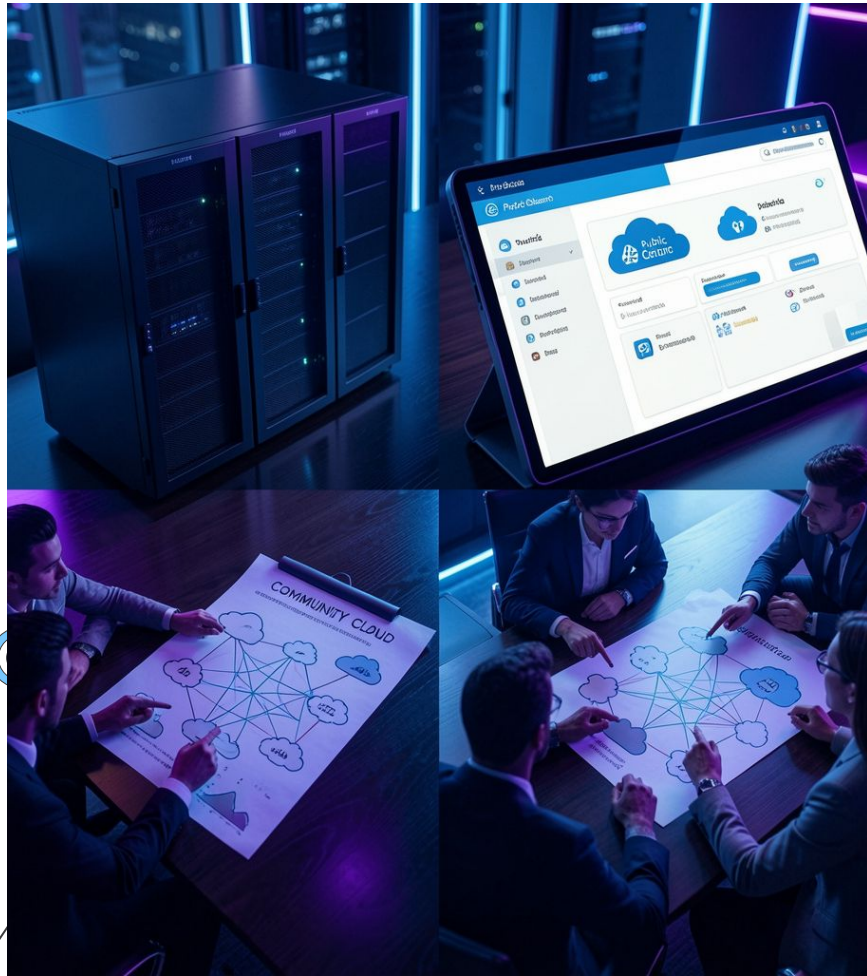
Infrastructure as a Service (**IaaS**)

Platform as a Service (**PaaS**)

Software as a Service (**SaaS**)



Service Model	AWS	Azure	GCP
 Infrastructure as a Service (IaaS)	 Amazon EC2	 Azure Virtual Machine	 Compute Engine
 Platform as a Service (PaaS)	 AWS Elastic Beanstalk	 Azure App Service	 Google App Engine
 Software as a Service (SaaS)	 Amazon WorkDocs	 Microsoft 365	 Google Workspace



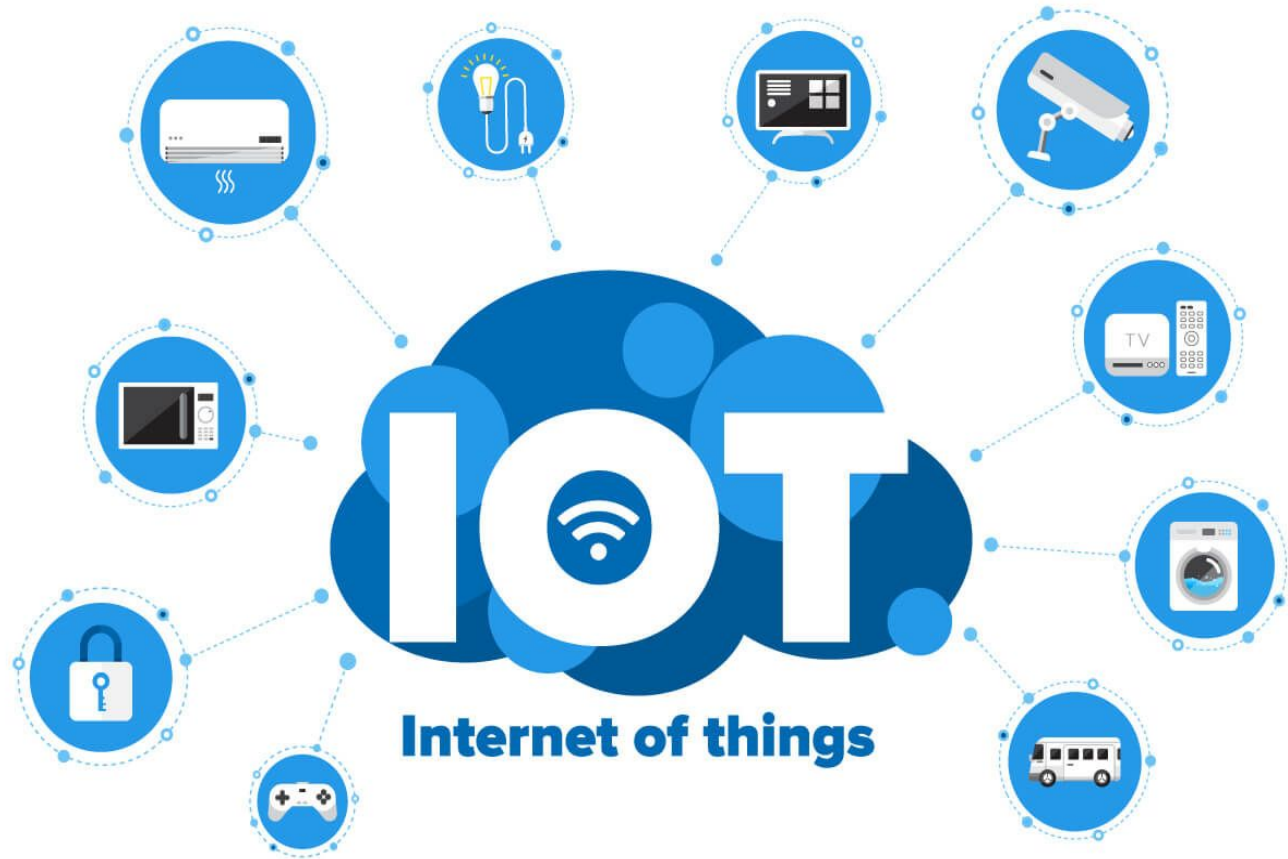
07

IoT in Cloud Computing and everyday use

What is IoT?

Internet of things:

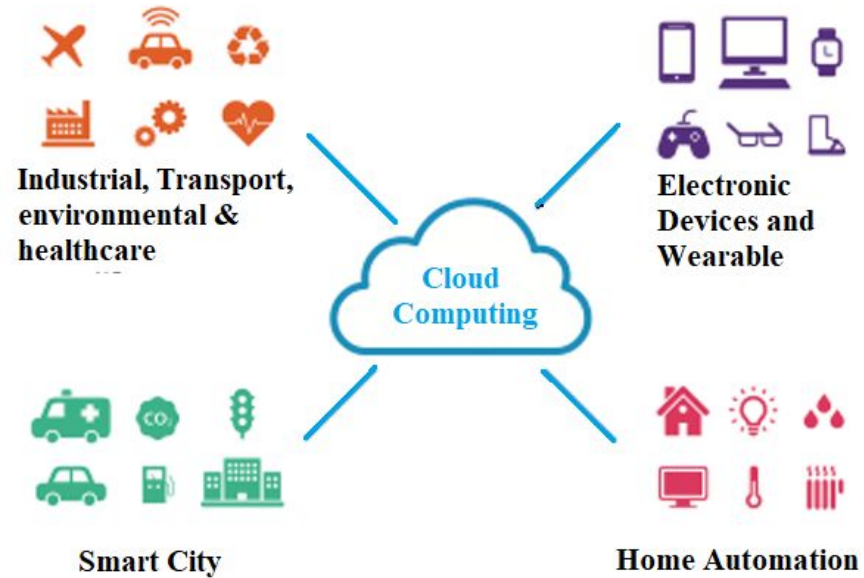
- receive
- collect
- send

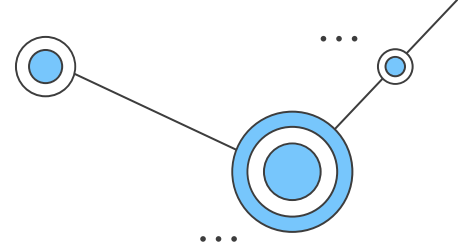
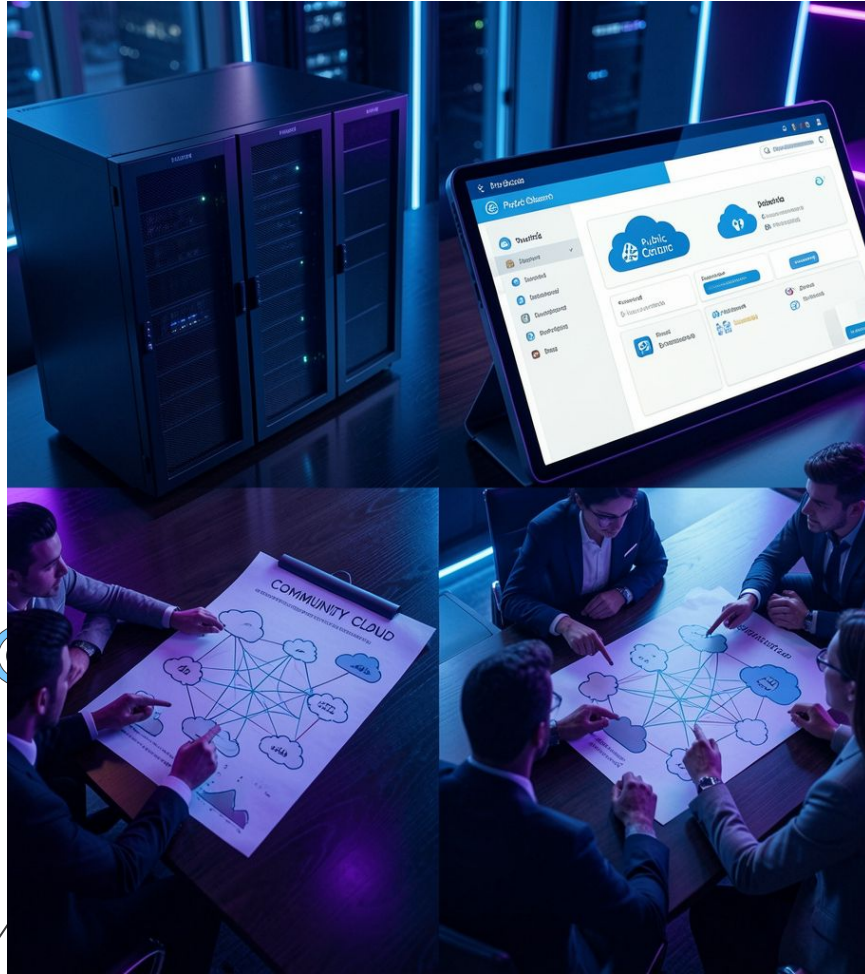


IoT and Cloud Computing and everyday use

- Cloud Services
- Cloud store
- Analyze data
- Allow remote control
- Monitoring in real time

Integration of Cloud Computing and Internet of Things





08

How companies use and benefit from Cloud Services

Cloud Services in companies

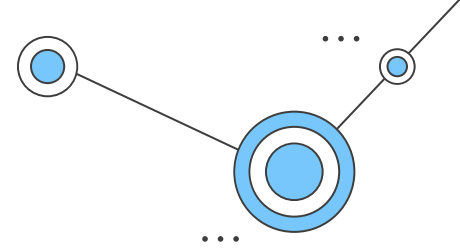
- Stock online
- Remote work
- Communication
- Backup and security

Benefits from Cloud services

- Cost reduction
- Scalability
- Greater productivity
- Remote access



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https://tryhackme.com/?utm_source=chatgpt.com

